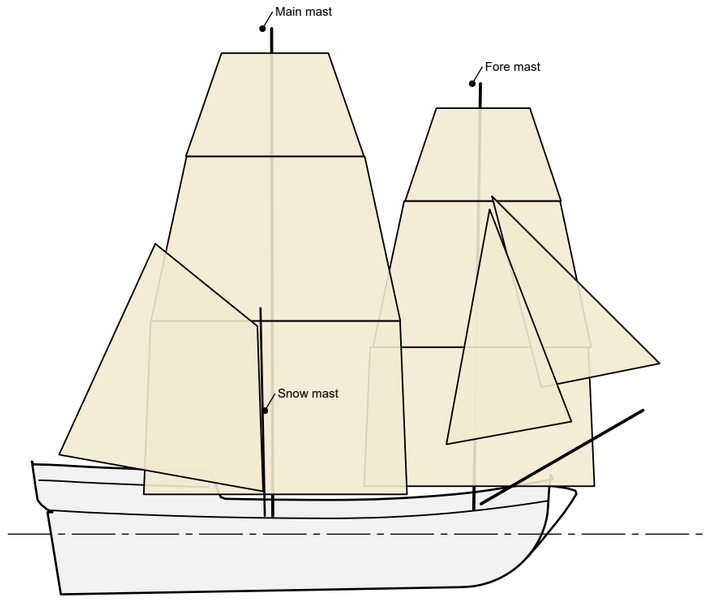
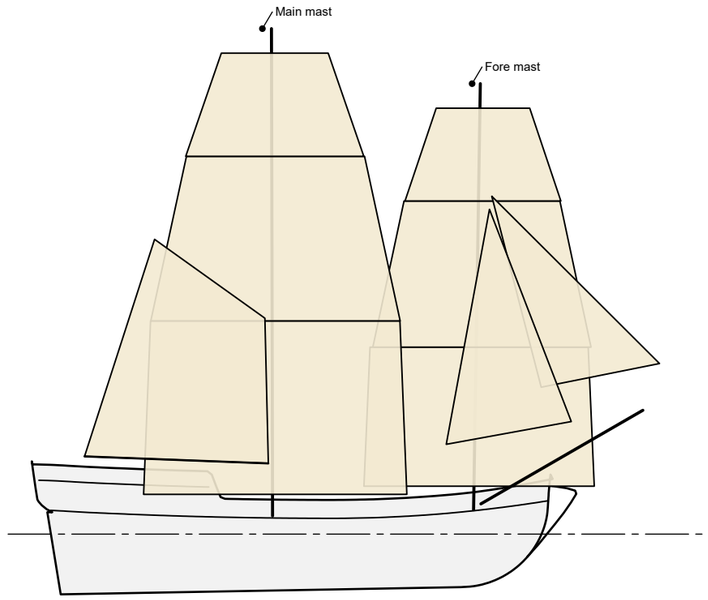
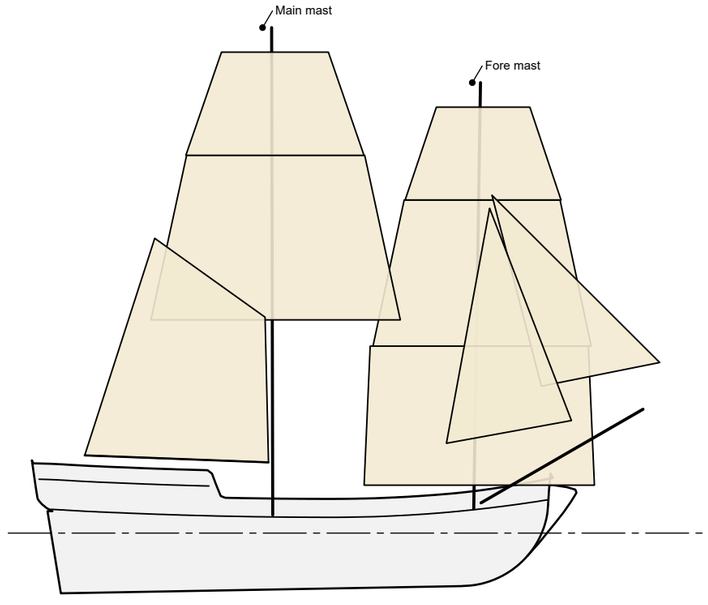
Snow (reference model, about 1710)



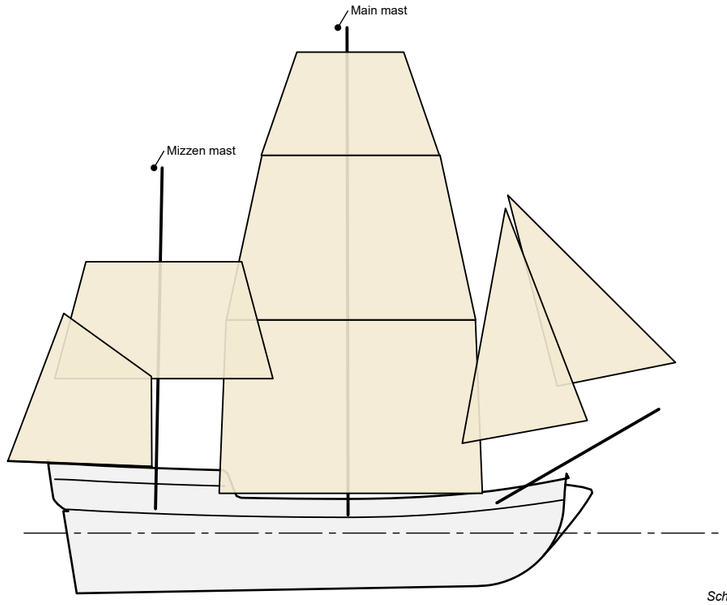
Later brig (gaff sail on the main mast, boom)



Brigantine (no main course)



Ketch (main and mizzen)



Schematic: same hull and spar proportions, sails set; no dimensions.

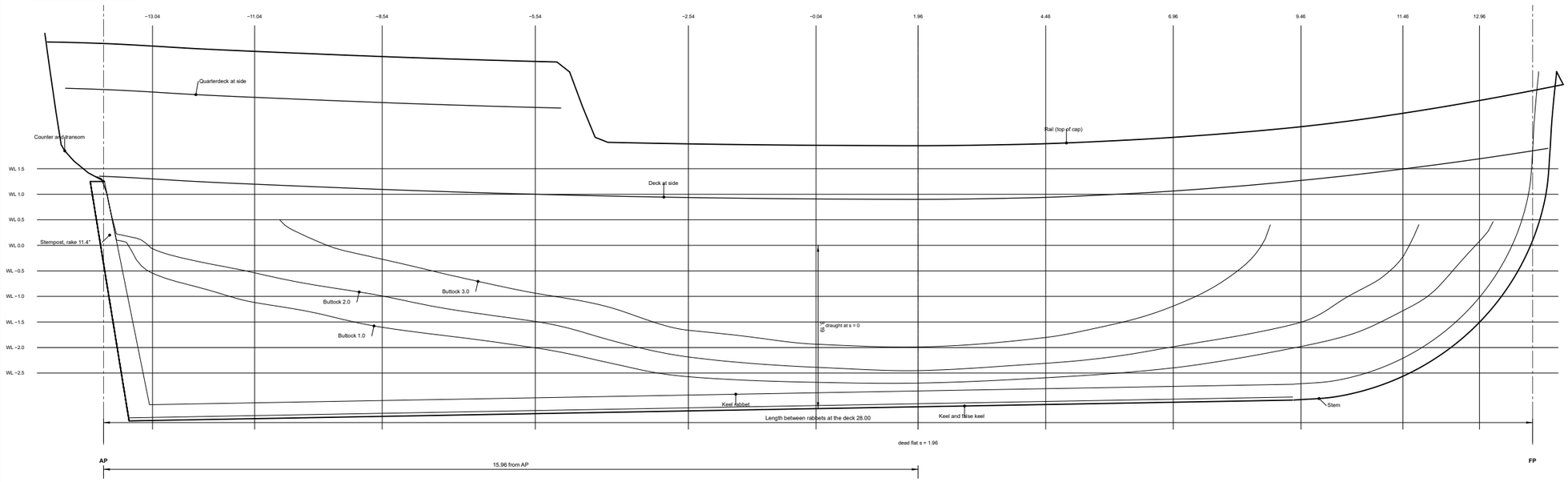
Rigs on one hull

Snow, later brig, brigantine and ketch on the hull of the reference model

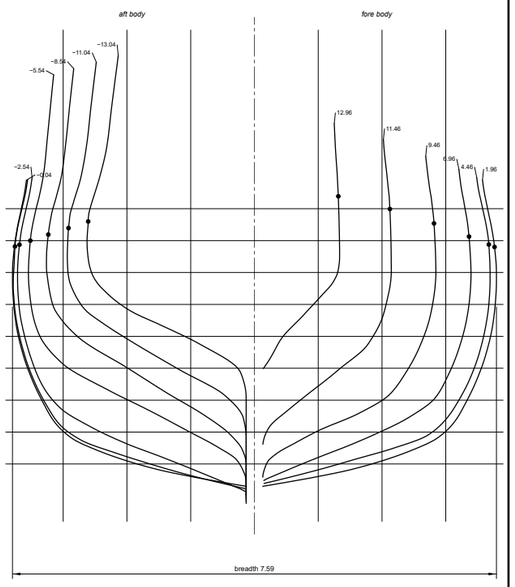
Scale	Sheet	Format
1:350	1 / 19	A3, dimensions in metres

Reference model - snow, about 1710 - drawn by script from the model parameters

Sheer plan (profile) 1:48



Body plan 1:48



Half-breadth plan 1:48

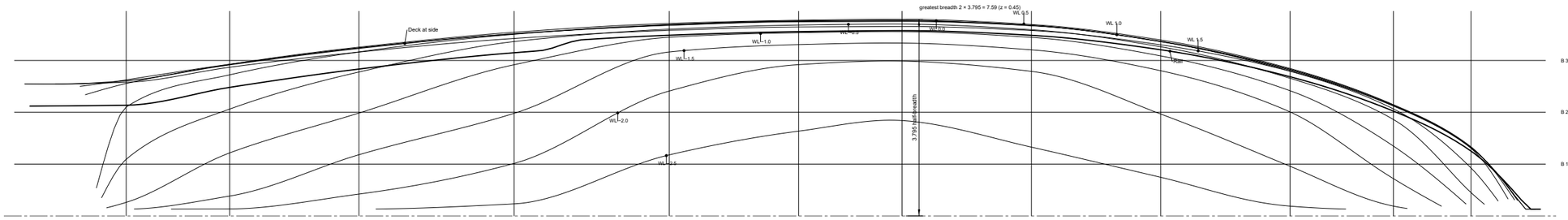


Table of offsets (research table, moulded, m)

s	z_df	keel	rabbet	WL −2.5	WL −2.0	WL −1.5	WL −1.0	WL −0.5	WL 0.0	WL 0.5	deck z	deck y	b	h_b	rail z	rail y
−13.04	−15.00	−3.37	−3.12	0.13	0.13	0.13	0.22	1.03	2.04	2.49	1.30	2.54	2.56	0.88	2.35	2.27
−11.04	−13.00	−3.33	−3.08	0.13	0.13	0.32	1.15	1.99	2.66	2.86	1.20	2.85	2.87	0.75	2.25	2.60
−8.54	−10.50	−3.29	−3.04	0.13	0.35	1.13	1.92	2.71	3.10	3.19	1.09	3.17	3.20	0.61	2.14	2.94
−5.54	−7.50	−3.23	−2.98	0.20	0.95	1.90	2.86	3.30	3.44	3.47	1.00	3.45	3.48	0.48	2.05	3.24
−2.54	−4.50	−3.18	−2.93	1.11	2.35	3.11	3.38	3.54	3.63	3.65	0.94	3.62	3.65	0.40	1.99	3.42
−0.04	−2.00	−3.14	−2.89	1.57	2.87	3.24	3.47	3.62	3.70	3.71	0.91	3.69	3.72	0.36	1.96	3.49
1.96	0.00	−3.10	−2.85	1.78	2.93	3.27	3.49	3.64	3.71	3.73	0.90	3.70	3.73	0.35	1.95	3.51
4.46	2.50	−3.06	−2.81	1.26	2.73	3.14	3.38	3.53	3.61	3.63	0.94	3.61	3.63	0.37	1.99	3.42
6.96	5.00	−3.01	−2.76	0.68	1.90	2.74	3.02	3.20	3.30	3.33	1.06	3.30	3.33	0.43	2.11	3.14
9.46	7.50	−2.97	−2.72	0.17	0.90	1.96	2.35	2.59	2.73	2.78	1.27	2.75	2.78	0.52	2.32	2.62
11.46	9.50	–	−2.21	–	0.17	0.66	1.28	1.81	2.00	2.08	1.49	2.06	2.08	0.63	2.54	1.96
12.96	11.00	–	−1.02	–	–	–	0.13	0.39	0.86	1.24	1.69	1.25	1.27	0.73	2.74	1.19

Generator table (outside planking, lines parallel to the keel): data/offsets.csv

1:48

0 1 2 3 4 5 6 7 8 9 10 m

On A3 the drawing is at 1:96 - use the scale bar.
Sheer plan and body plan: z above the design waterline; half-breadth plan: x from the centreline.
Lines as generated from the table; the run below WL −1.0 is not faired (bumps of 1-5 cm), see guide chapter 02.

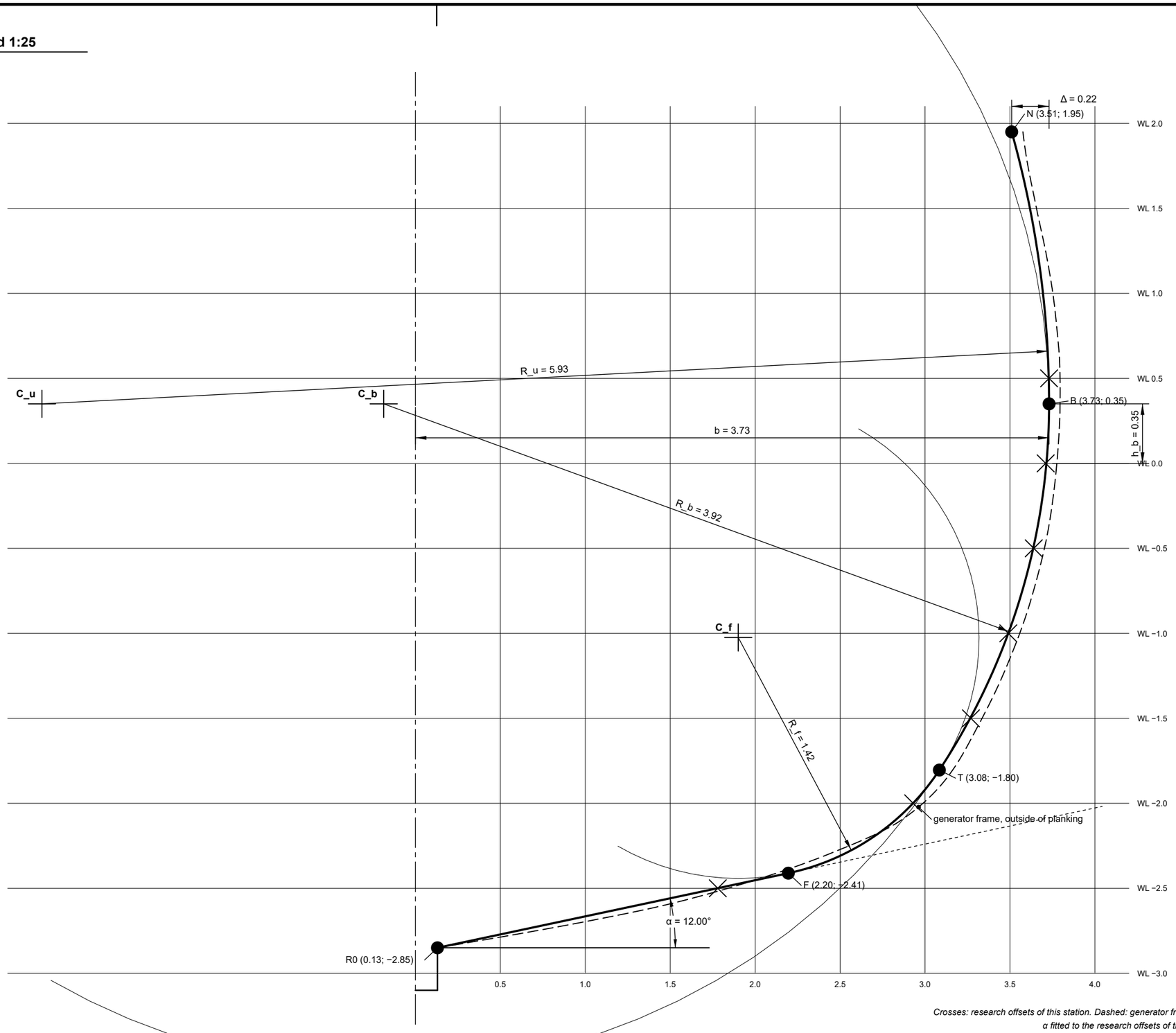
Lines plan

Stations, waterlines every 0.5 m, buttocks 1, 2, 3 m; lines to the outside of the planking (generator surface)

Scale	Sheet	Format
1:48 (A1)	2 / 19	A1, dimensions in metres

Reference model - snow, about 1710 - drawn by script from the model parameters

Midship frame (s = +1.96) drawn with arcs, moulded 1:25



Crosses: research offsets of this station. Dashed: generator frame outside the planking.
α fitted to the research offsets of the station (least squares).

Arc data (m)

α	12.00°
R0	(0.13; −2.85)
F	(2.20; −2.41)
R_f = 0.38 b	1.42
C_f	(1.90; −1.02)
R_b = 1.05 b	3.92
C_b	(−0.19; 0.35)
T	(3.08; −1.80)
R_u	5.93
C_u	(−2.20; 0.35)
rms vs. table	0.003

1:25

0 0.4 0.8 1.2 1.6 2 m

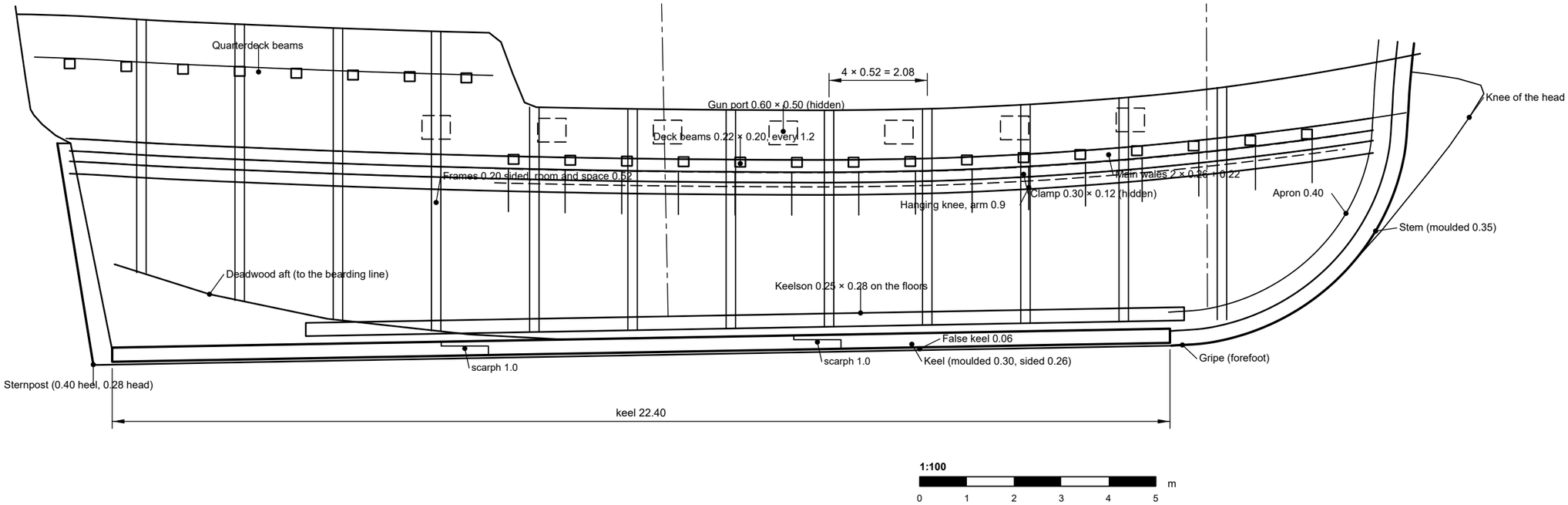
Frame construction by arcs

Deadrise line, floor sweep, breadth sweep, top-timber sweep (chapter 02)

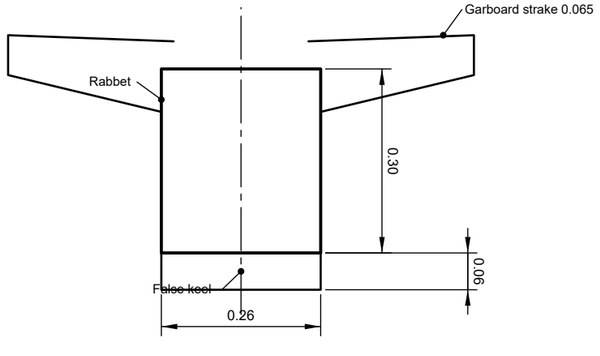
Scale	Sheet	Format
1:25	3 / 19	A3, dimensions in metres

Reference model - snow, about 1710 - drawn by script from the model parameters

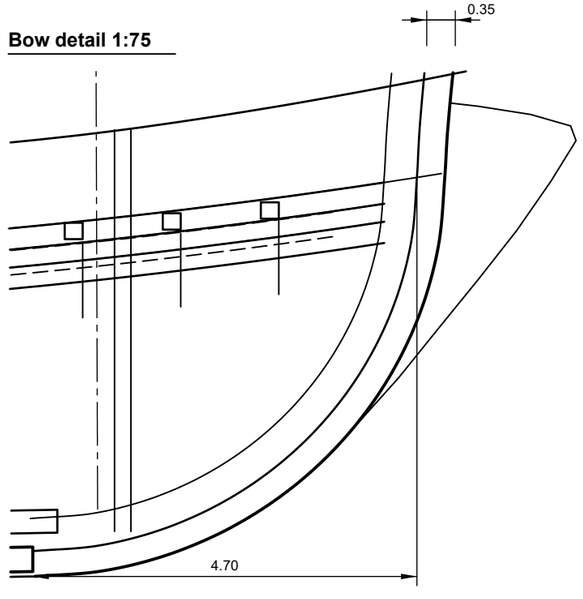
Structural profile (planking removed, one frame in four) 1:100



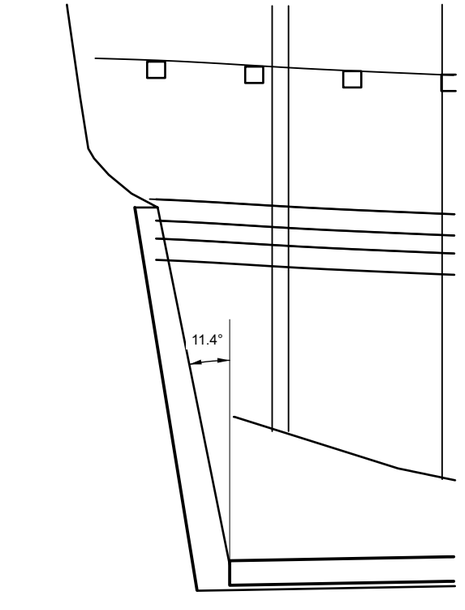
Keel section 1:10



Bow detail 1:75



Stern detail 1:75



Members not built in the 3D model (keelson, frames, beams, knees, apron) use the proportions of chapter 03 (our inference).

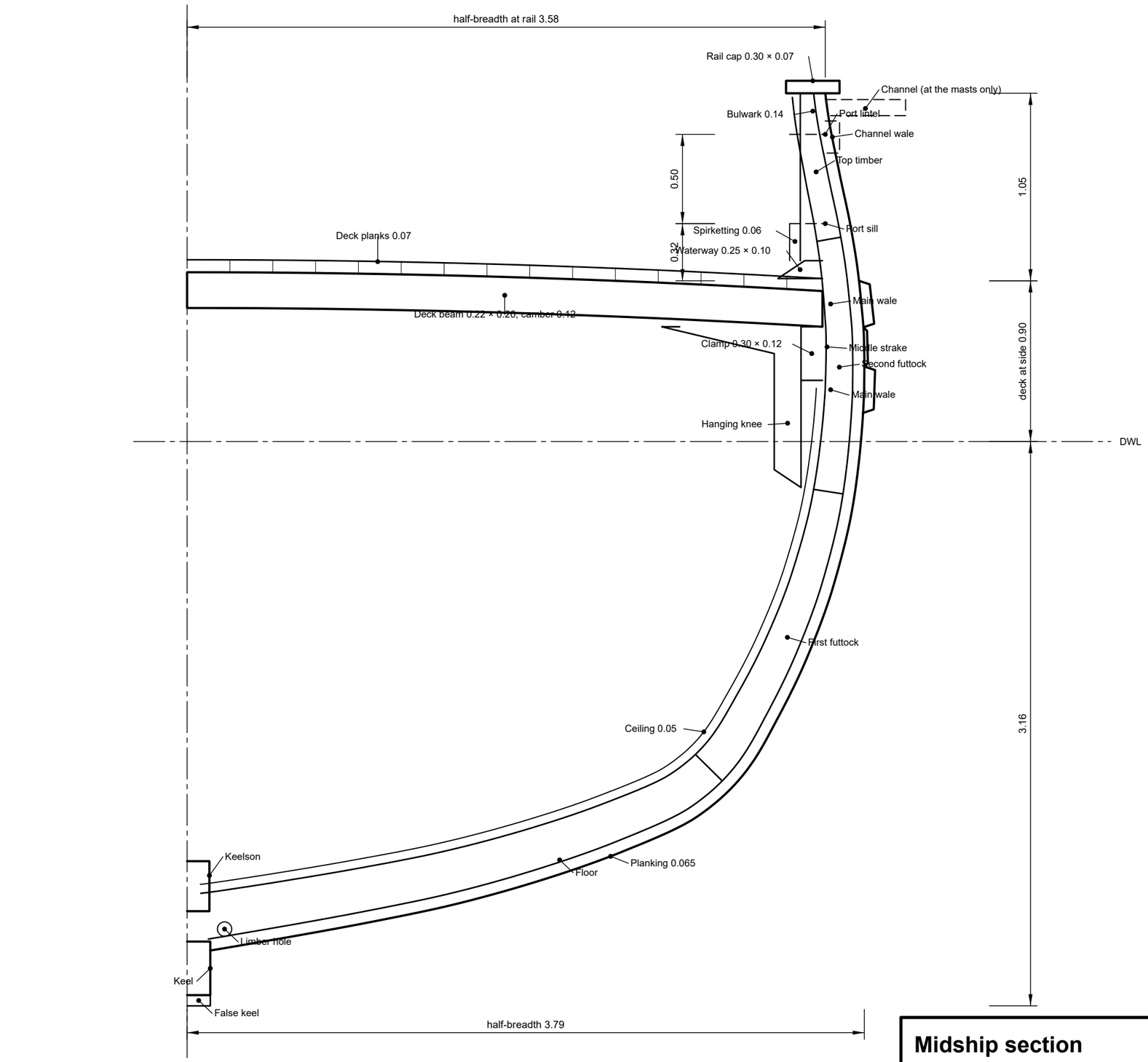
Hull structure - profile

Backbone, frames, wales, clamp, beams and knees

Scale	Sheet	Format
1:100, 1:75, 1:10	4 / 19	A3, dimensions in metres

Reference model - snow, about 1710 - drawn by script from the model parameters

Midship section s = +1.96, port half (looking forward) 1:25



Keelson, frames, ceiling, clamp, beams and knees: proportions of chapter 03 (our inference); hull surface, wales, deck, bulwark, rail and ports from the model parameters.

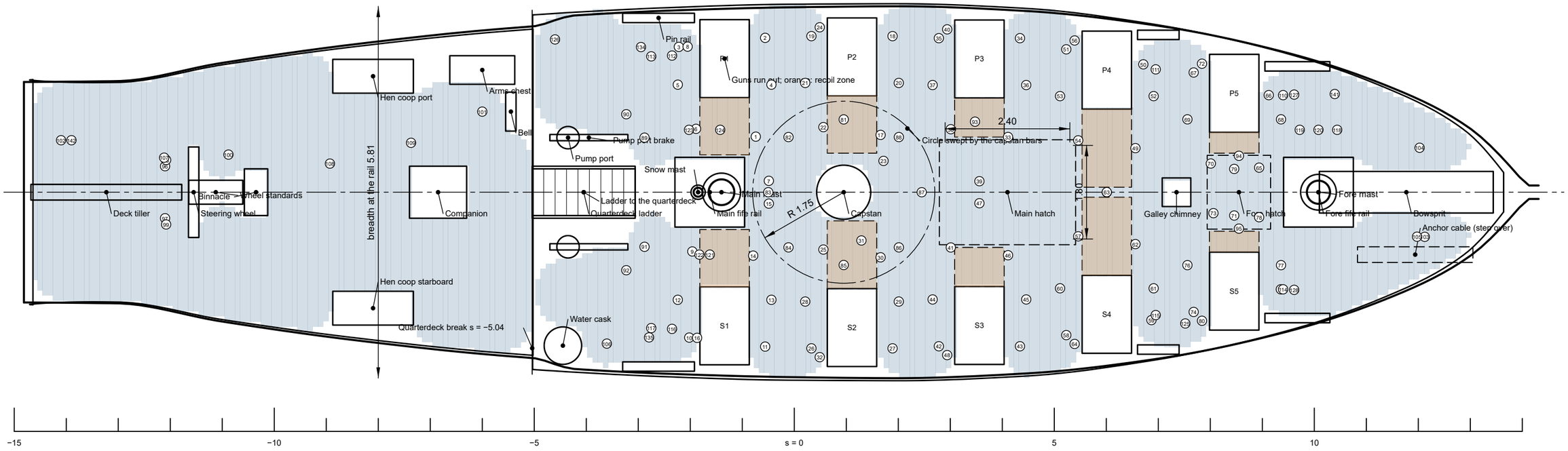
Midship section

Structure at the dead flat; planking outside the generator surface

Scale	Sheet	Format
1:25	5 / 19	A3, dimensions in metres

Reference model - snow, about 1710 - drawn by script from the model parameters

Deck plan: main deck and quarterdeck 1:80



Work stations (numbers on the plan; data/stations.csv)

gun captain: 1, 9, 17, 25, 33, 41…
gun loader: 2, 10, 18, 26, 34, 42…
gun sponger: 3, 11, 19, 27, 35, 43…
gun side tackle: 4, 5, 12, 13, 20, 21…
gun train tackle: 6, 14, 22, 30, 38, 46…
gun powder boy: 7, 15, 23, 31, 39, 47…
port lid: 8, 16, 24, 32, 40, 48…
capstan: 81–88
pump: 89–92
hatch: 93–95

wheel: 96–97
rudder: 98–99
compass: 100
bell: 101
boat: 102
anchor: 103
cathead: 104–105
water cask: 106
braces: 107, 112, 116
vangs: 108

sheets: 109, 111, 115, 119, 123
clewlines: 110, 113, 114, 117, 118, 122
lifts: 120, 124
halliards: 121, 125, 126
shrouds climb: 127, 128, 134, 135
top: 129, 136
yardarm: 130, 131, 132, 133, 137, 138…
hand lead: 141
log line: 142

walkways at least 0.8 m wide (from the walking test of the model)

recoil zone of each gun (to the first obstacle)

1:80

0 1 2 3 4 5 m

Stations aloft (tops, yardarms, shrouds, channels) are listed but not drawn.
Dashed outlines: obstacles one can step over (cable, thresholds).
Guns labelled P (port) / S (starboard) and number from forward.

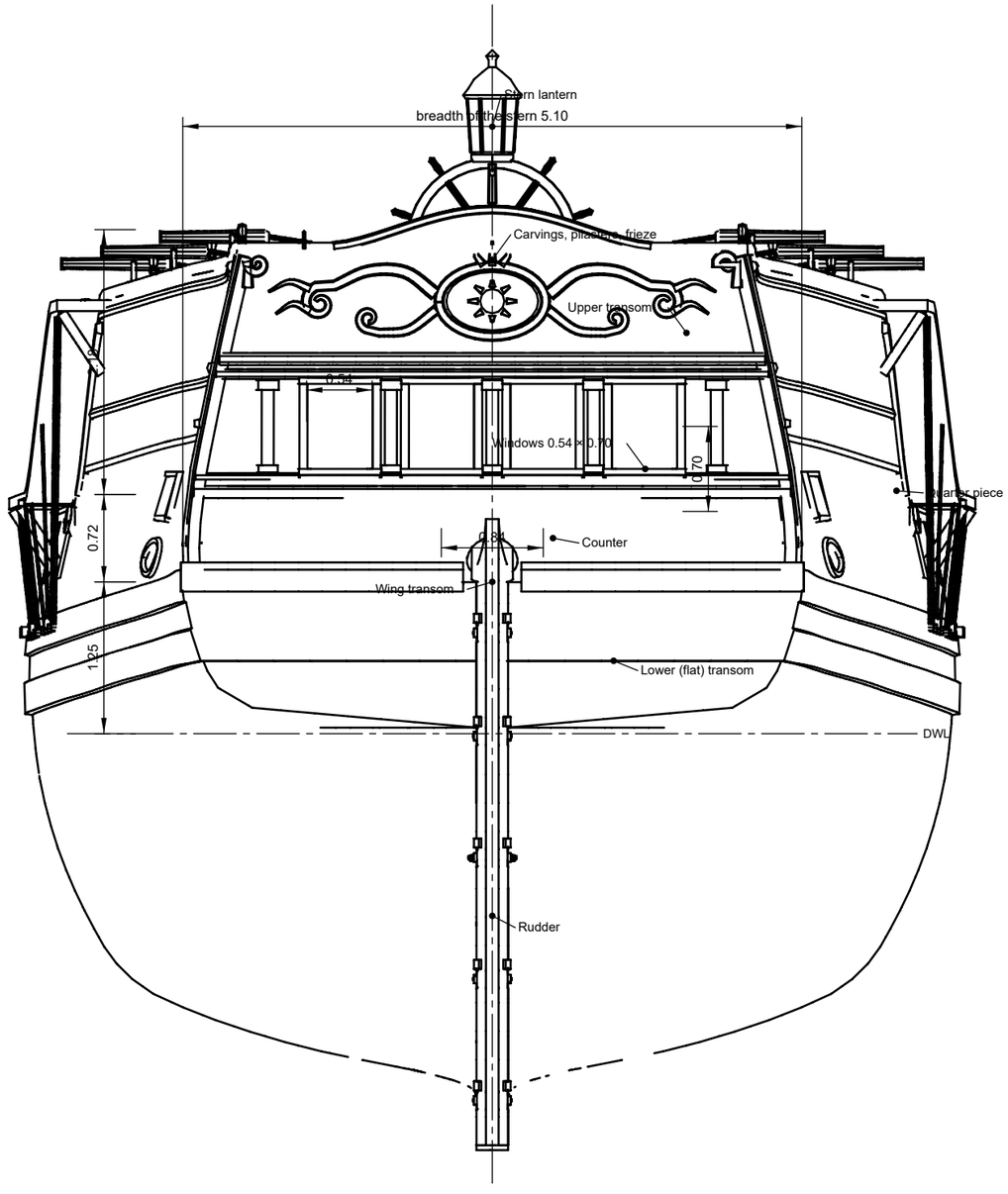
Deck plan

Main deck and quarterdeck with fittings, walkways, recoil zones and work stations

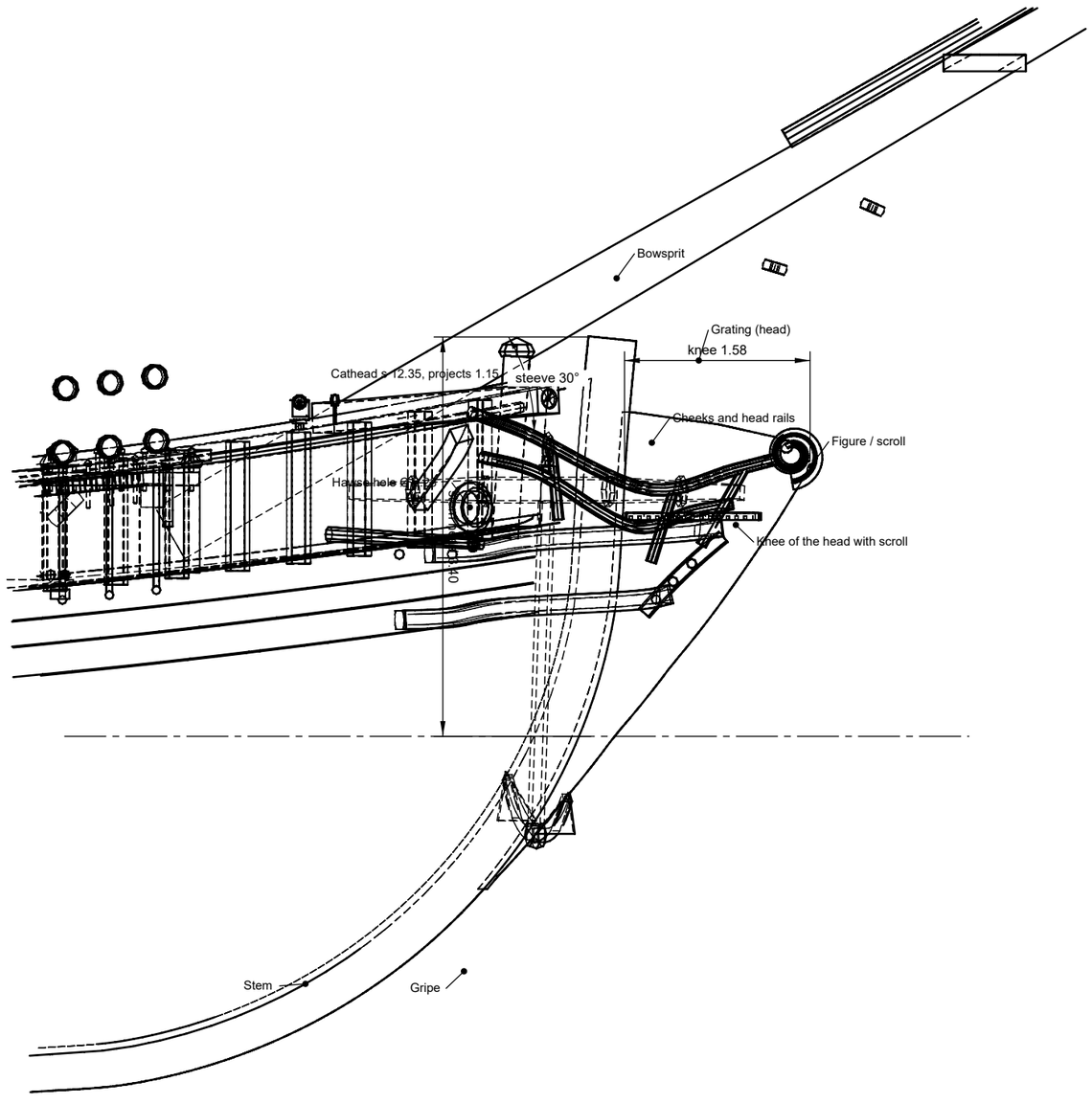
Scale	Sheet	Format
1:80	6 / 19	A3, dimensions in metres

Reference model - snow, about 1710 - drawn by script from the model parameters

Stern seen from aft 1:50



Head from starboard 1:50



1:50

0 1 2 3 4 5 m

Visible edges solid, hidden edges dashed (ray test against all meshes); in the stern view hidden edges are left out for legibility.

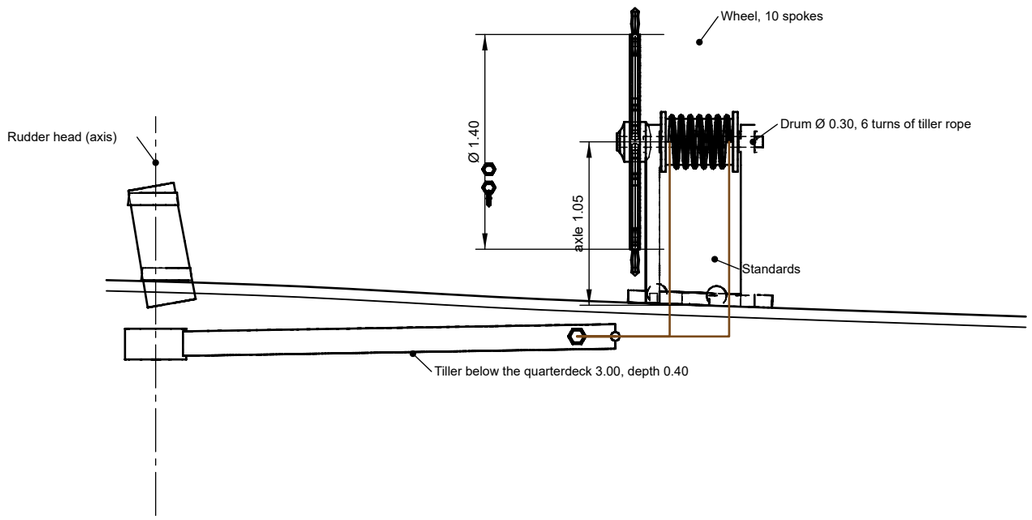
Stern and head

Projected from the 3D model with hidden lines removed; dimensions from the parameters

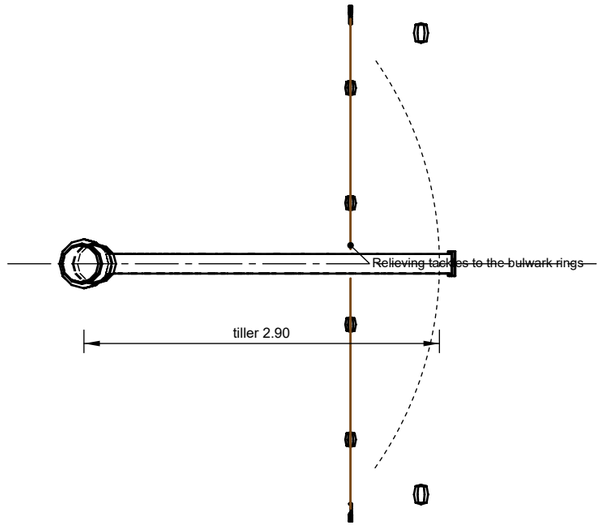
Scale	Sheet	Format
1:50	7 / 19	A3, dimensions in metres

Reference model - snow, about 1710 - drawn by script from the model parameters

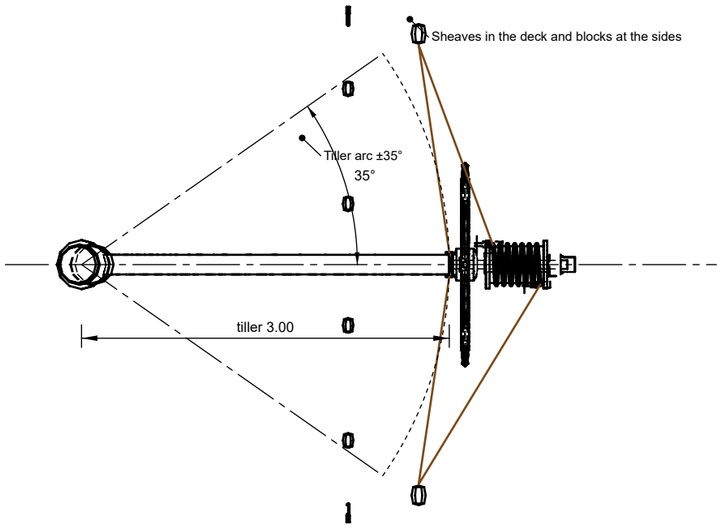
Wheel variant, side view from starboard 1:40



Deck tiller variant, plan 1:50



Wheel variant, plan 1:50



Gear ratio of the wheel

n = L · sin 35° / (π · D)

= 3.00 · 0.574 / (π · 0.30) = 1.83 turns

L - tiller length, D - drum diameter; turns of the wheel for 35° of rudder.

Tiller rope Ø 26 mm

1:40

0 0.4 0.8 1.2 1.6 2 m

1:50

0 0.4 0.8 1.2 1.6 2 m

Fittings projected from the 3D model; tiller ropes from the rigging data.

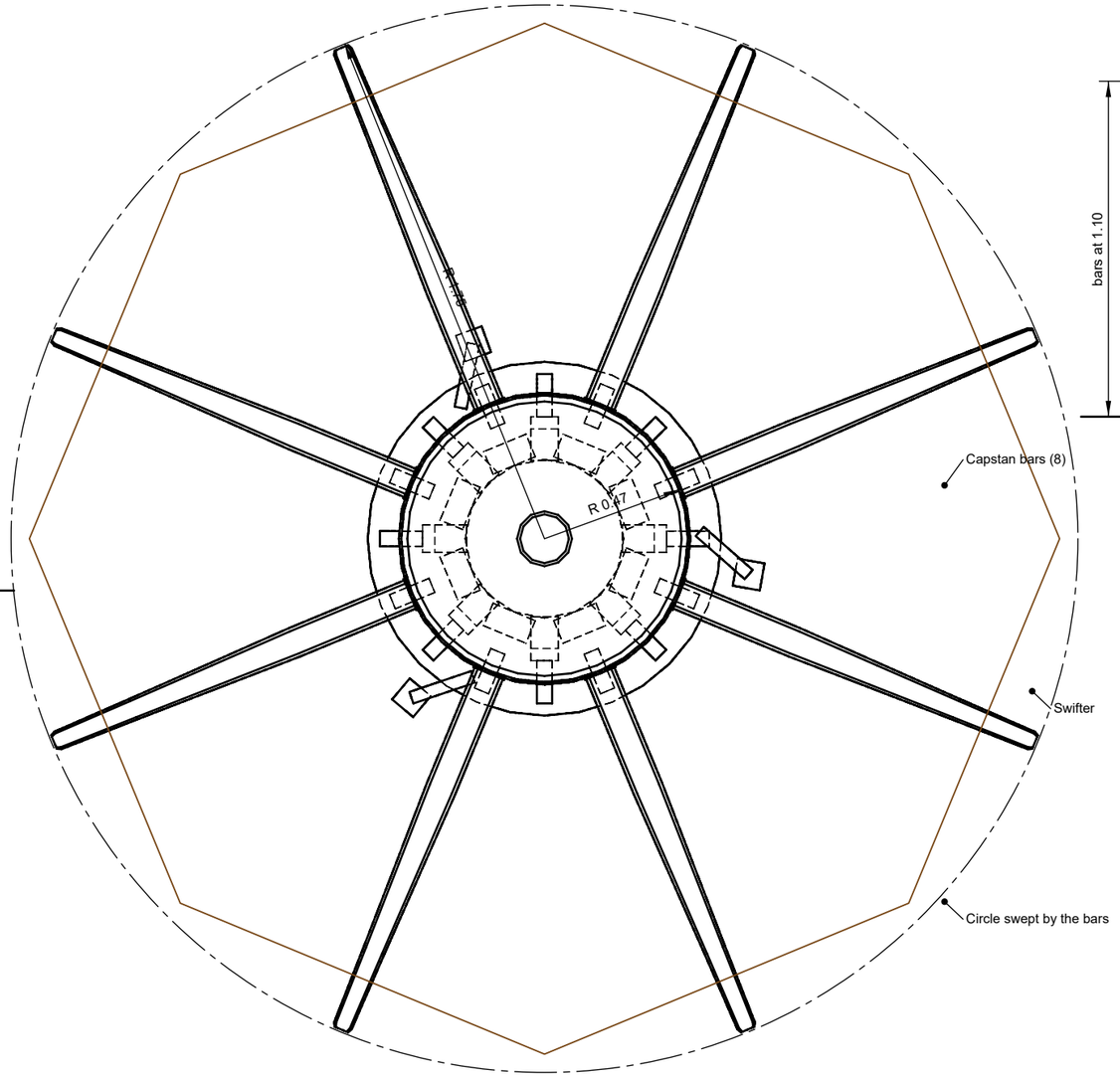
Steering gear

Wheel, drum, tiller rope, tiller; deck tiller variant

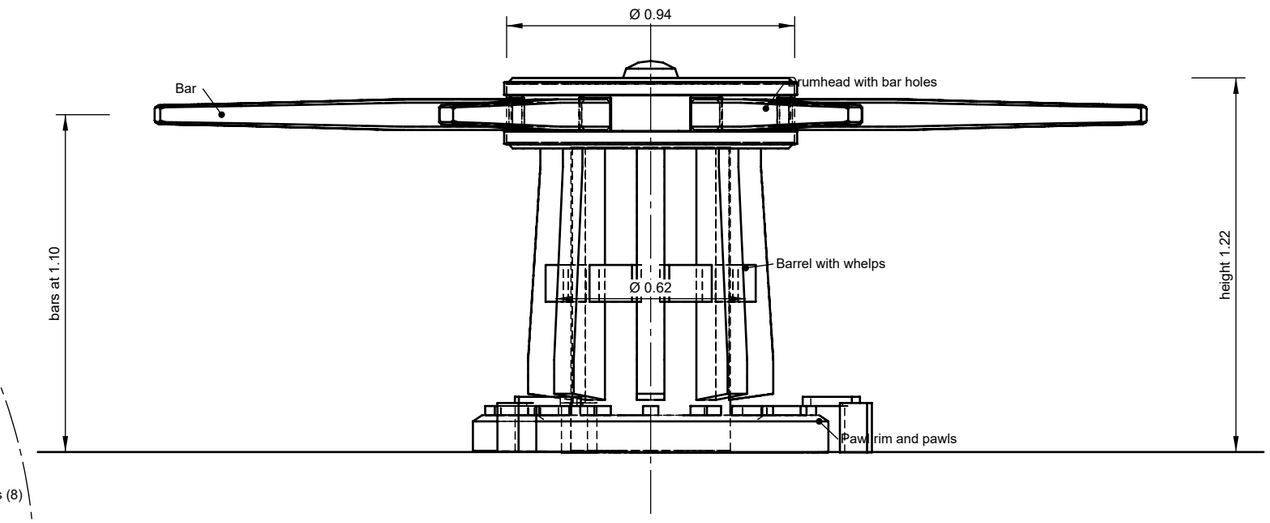
Scale	Sheet	Format
1:40, plans 1:50	8 / 19	A3, dimensions in metres

Reference model - snow, about 1710 - drawn by script from the model parameters

Plan 1:20



Elevation from starboard 1:20



1:20

0 0.4 0.8 1.2 1.6 2 m

Projected from the 3D model; the barrel turns about its vertical axis without limit.

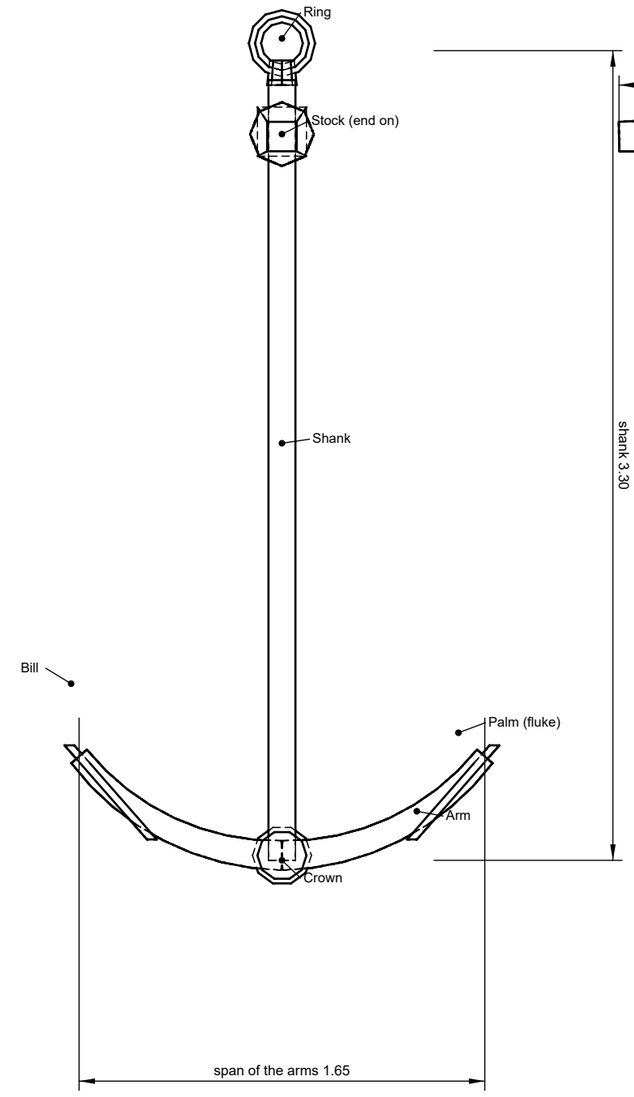
Capstan

Pawl rim, barrel with whelps, drumhead, bars, swifter

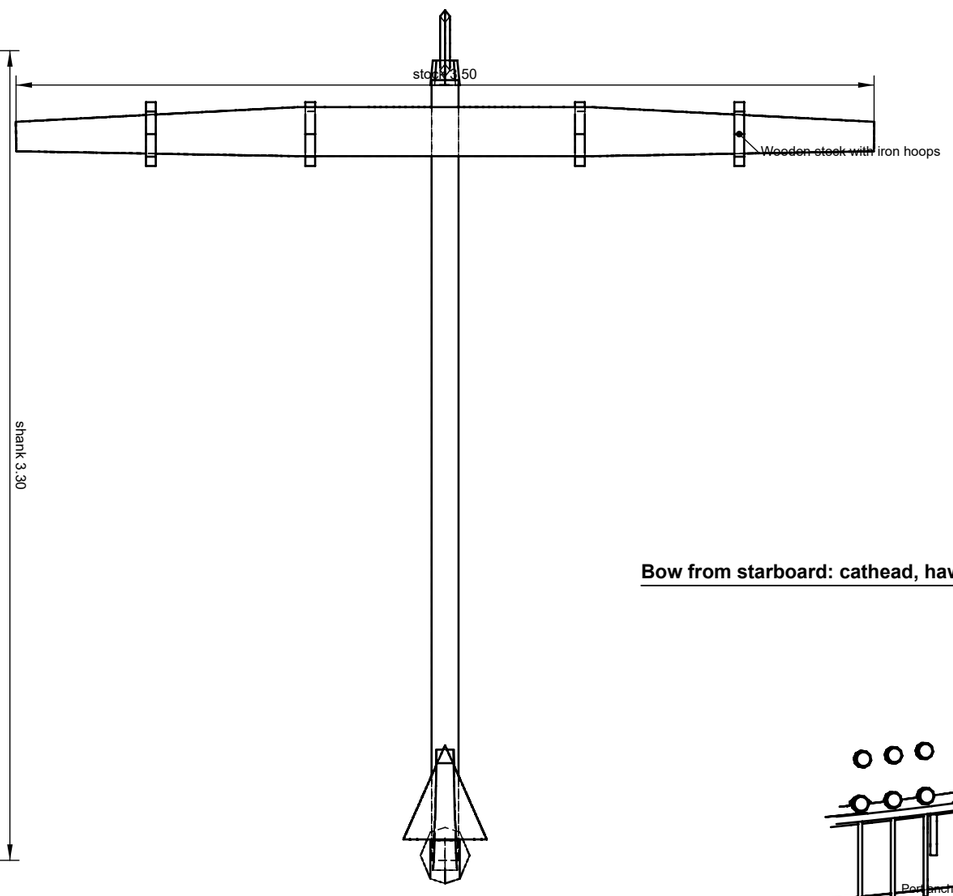
Scale	Sheet	Format
1:20	9 / 19	A3, dimensions in metres

Reference model - snow, about 1710 - drawn by script from the model parameters

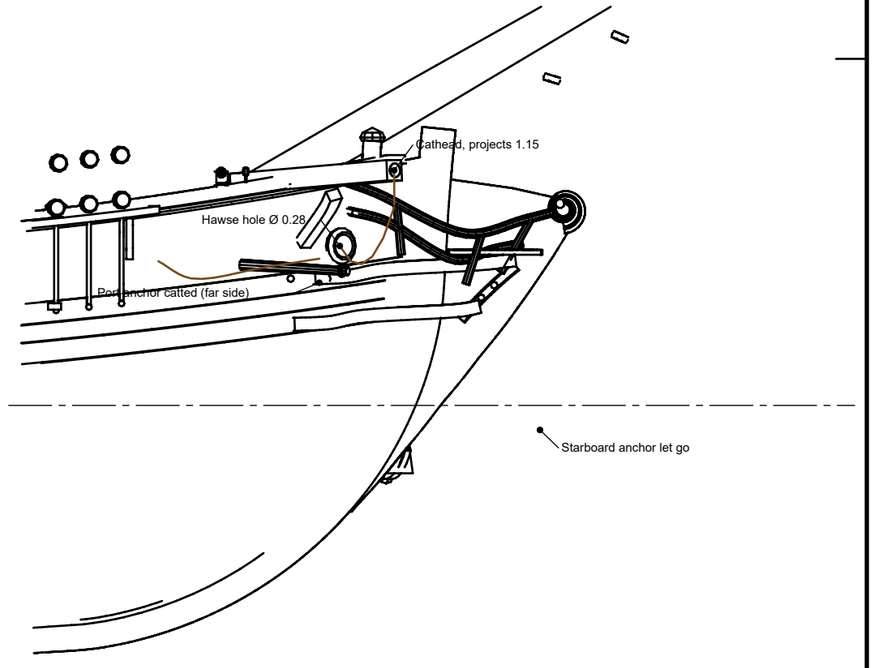
Bower anchor, arms in the plane of the drawing 1:25



Side view, stock across 1:25



Bow from starboard: cathead, hawse, cable 1:75



Anchor projected from the 3D model in its own coordinates; cable and cat fall from the rope data.

1:25

0 0.4 0.8 1.2 1.6 2 m

1:75

0 0.6 1.2 1.8 2.4 3 m

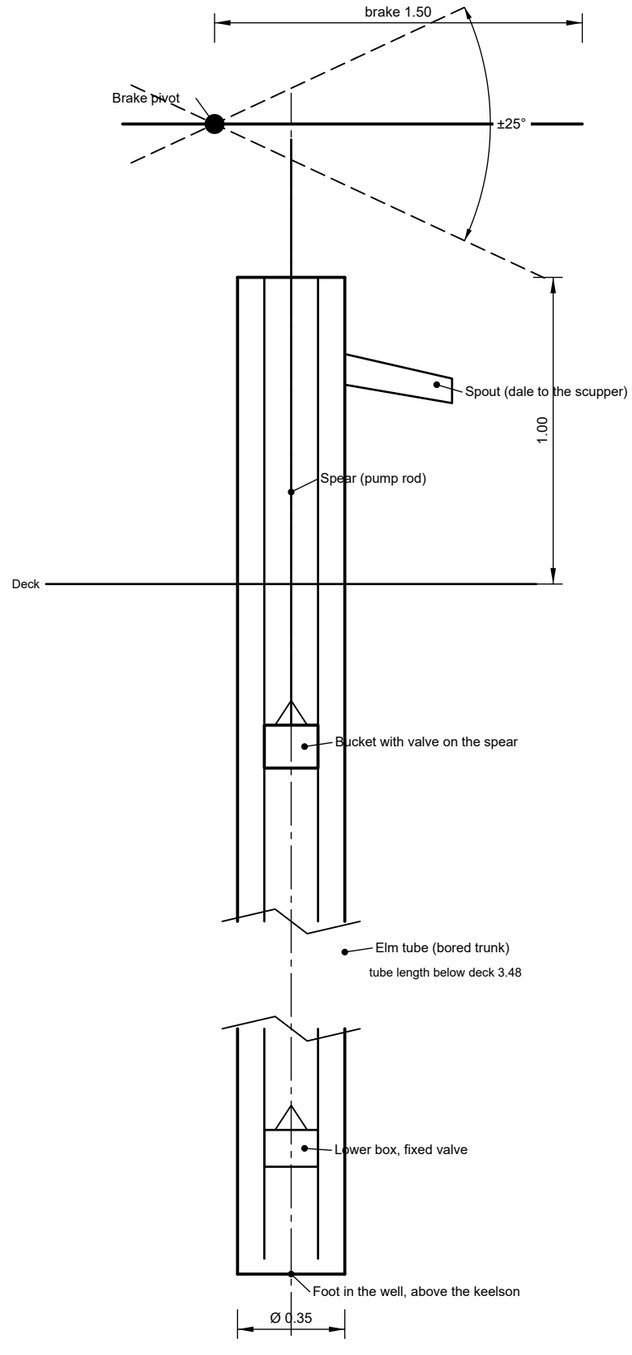
Anchor and cathead

Bower anchor with its parts; catted and let-go positions

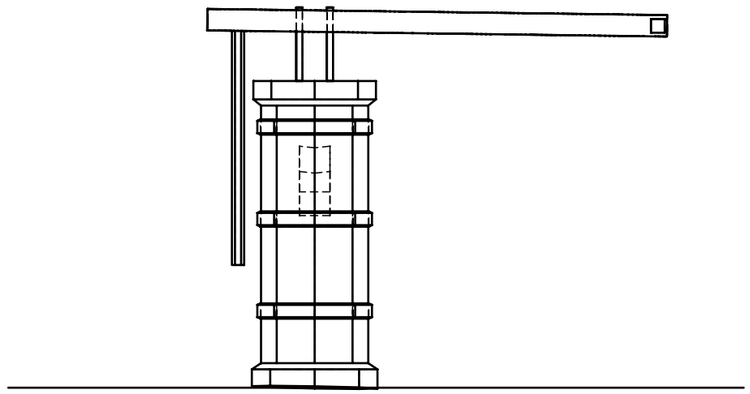
Scale	Sheet	Format
1:25, 1:75	10 / 19	A3, dimensions in metres

Reference model - snow, about 1710 - drawn by script from the model parameters

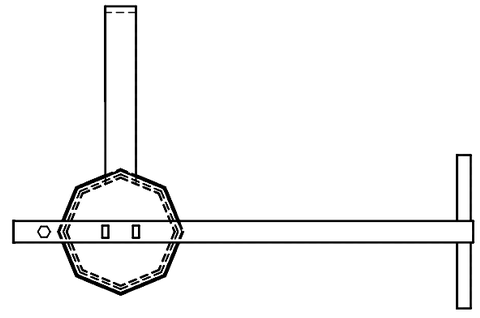
Section through the pump 1:20



Model pump from starboard (port pump) 1:20



Plan 1:20



1:20

0 0.4 0.8 1.2 1.6 2 m

Section drawn from the parameters (tube Ø, height, brake, swing); valves and tube length: our inference (chapter 09). The 3D model has only the outside.

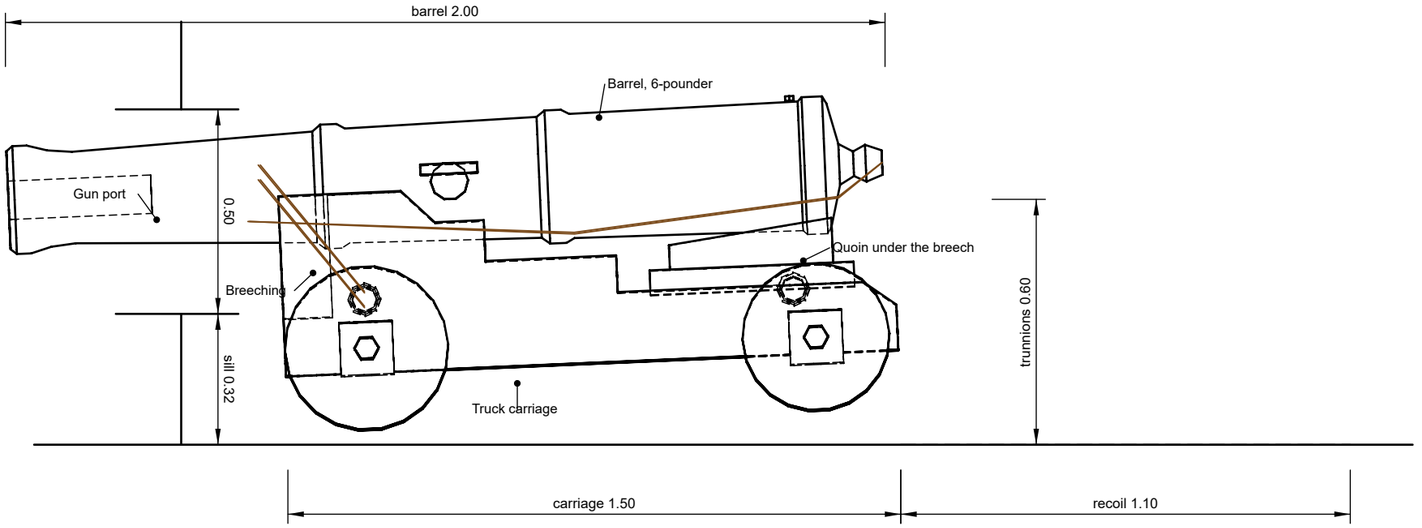
Pump

Elm-tree (common) pump: tube, bucket, valves, brake ±25°, spout

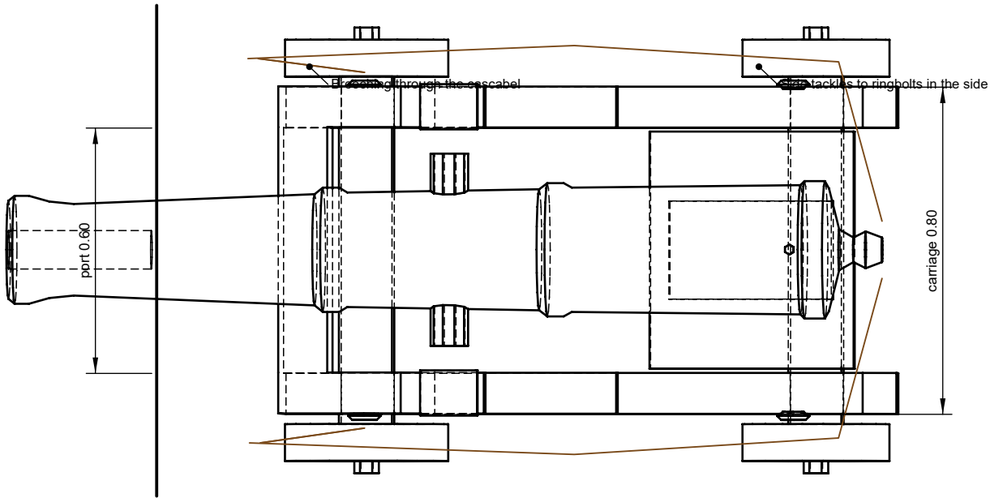
Scale	Sheet	Format
1:20	11 / 19	A3, dimensions in metres

Reference model - snow, about 1710 - drawn by script from the model parameters

Side elevation (looking forward), port gun run out 1:15



Plan 1:15



1:15

0 0.2 0.4 0.6 0.8 1 m

Gun, carriage and port projected from the 3D model; ropes from the rope data; recoil distance from the model's clearance test.

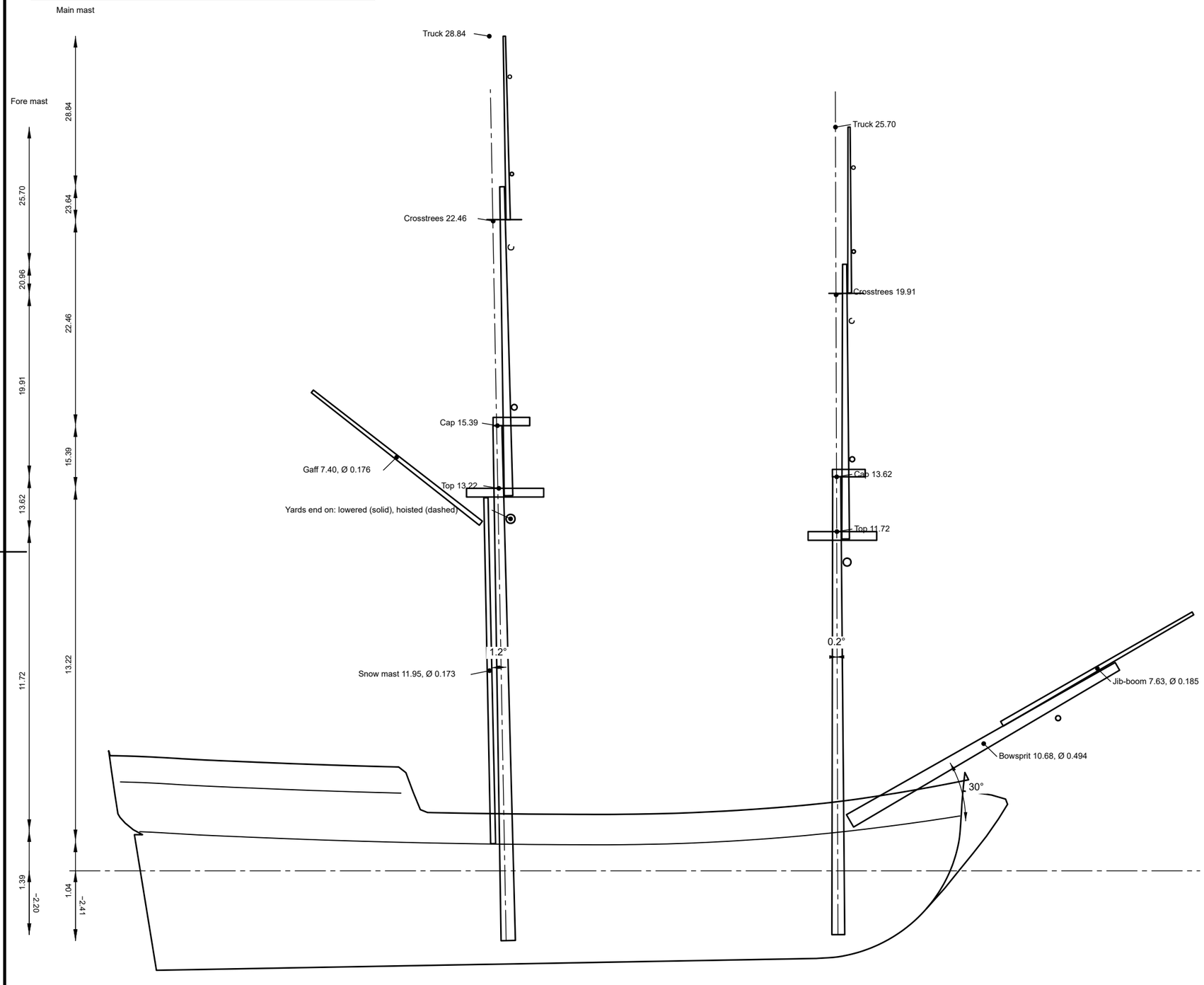
Gun and carriage

6-pounder on a truck carriage with breeching and side tackles

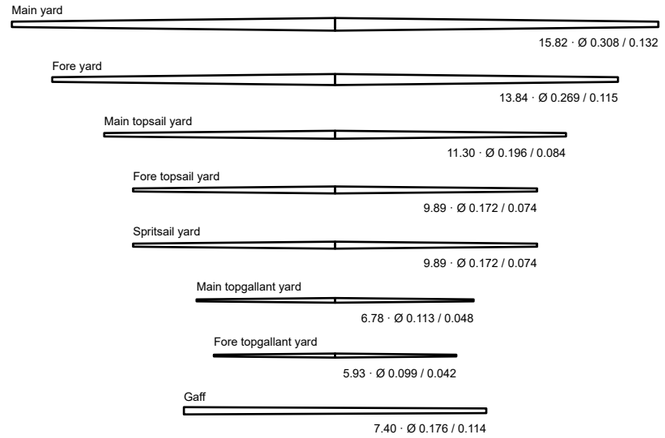
Scale	Sheet	Format
1:15	12 / 19	A3, dimensions in metres

Reference model - snow, about 1710 - drawn by script from the model parameters

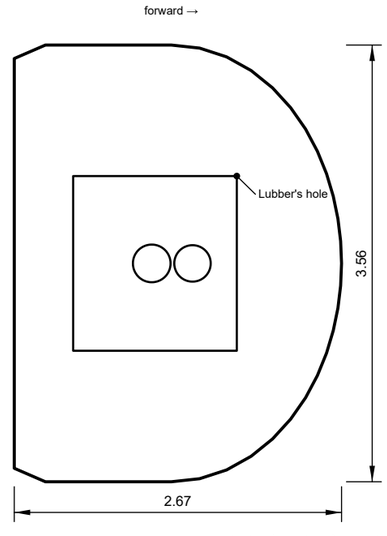
Spar plan, side elevation from starboard 1:150



Yards, gaff (length, diameter at slings / yardarm) 1:150



Main top and cap, plan 1:50



1:150

0 1 2 3 4 5 6 7 8 9 10 m

All lengths, diameters and heights: data/spars.csv.

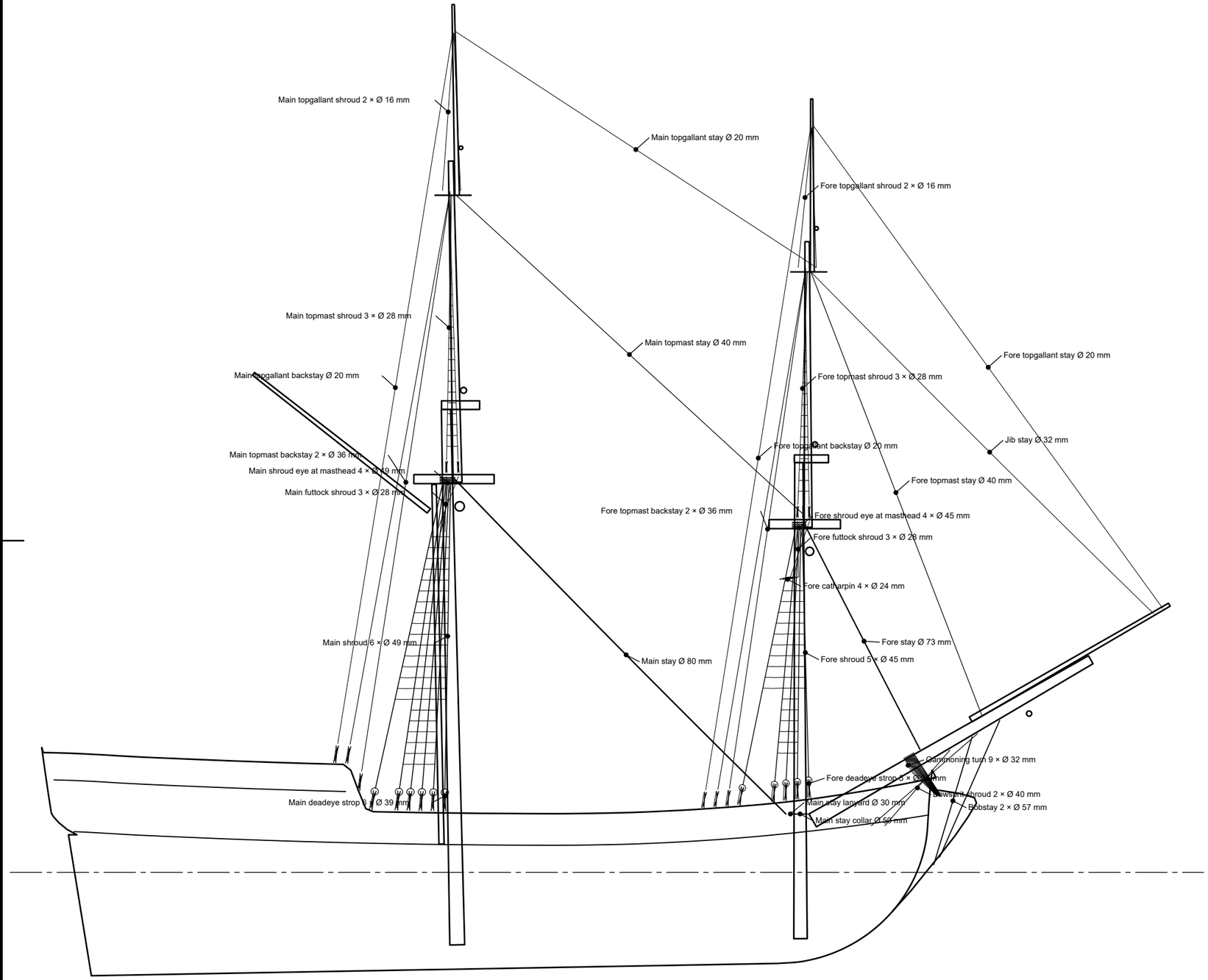
Spar plan

Masts with rake, heights, yards lowered and hoisted, bowsprit, jib-boom, gaff

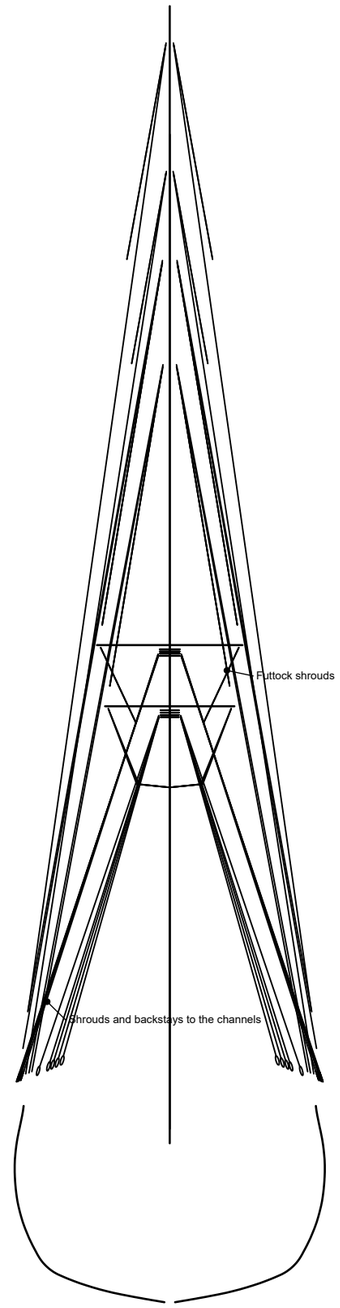
Scale	Sheet	Format
1:150, 1:50	13 / 19	A3, dimensions in metres

Reference model - snow, about 1710 - drawn by script from the model parameters

Side elevation from starboard 1:150



View from ahead 1:150



Each label: rope, number per side and diameter. Ratlines and lanyards thin.
Geometry and lengths: data/rigging-lines.json.

1:150

0 1 2 3 4 5 6 7 8 9 10 m

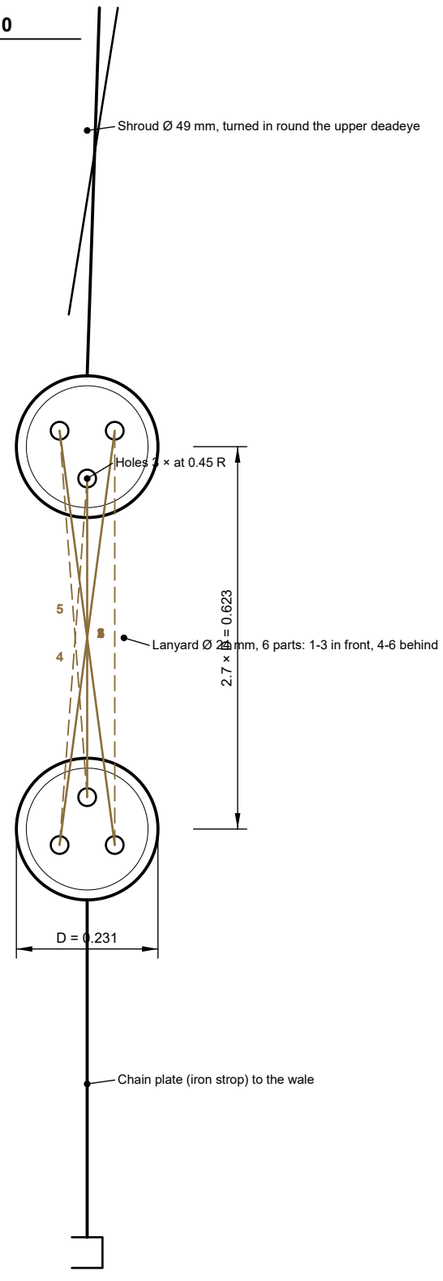
Standing rigging

Shrouds, topmast and topgallant shrouds, backstays, stays, bowsprit rigging

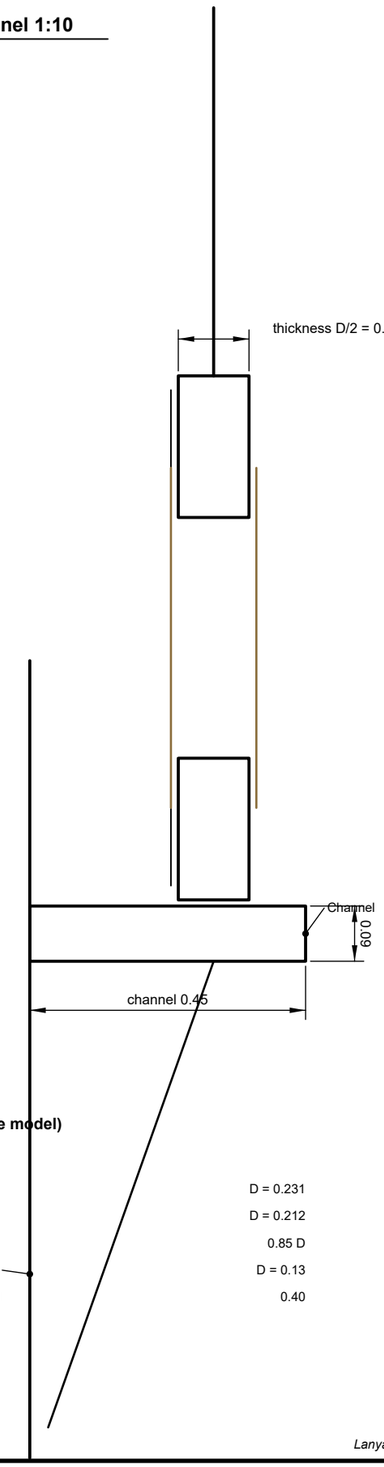
Scale	Sheet	Format
1:150	14 / 19	A3, dimensions in metres

Reference model - snow, about 1710 - drawn by script from the model parameters

Pair of deadeyes, front view (main shroud) 1:10



Side view with channel 1:10



Deadeye sizes (reference model)

D = 1.5 × 1 × π · d	
main shroud d = 49 mm	D = 0.231
fore shroud d = 45 mm	D = 0.212
backstays	0.85 D
topmast shrouds (inside tops)	D = 0.13
deadeye spacing on the channel	0.40

Lanyard rove as in the model: three parts in front, three behind; knot at the upper back hole.

1:10

0 0.2 0.4 0.6 0.8 1 m

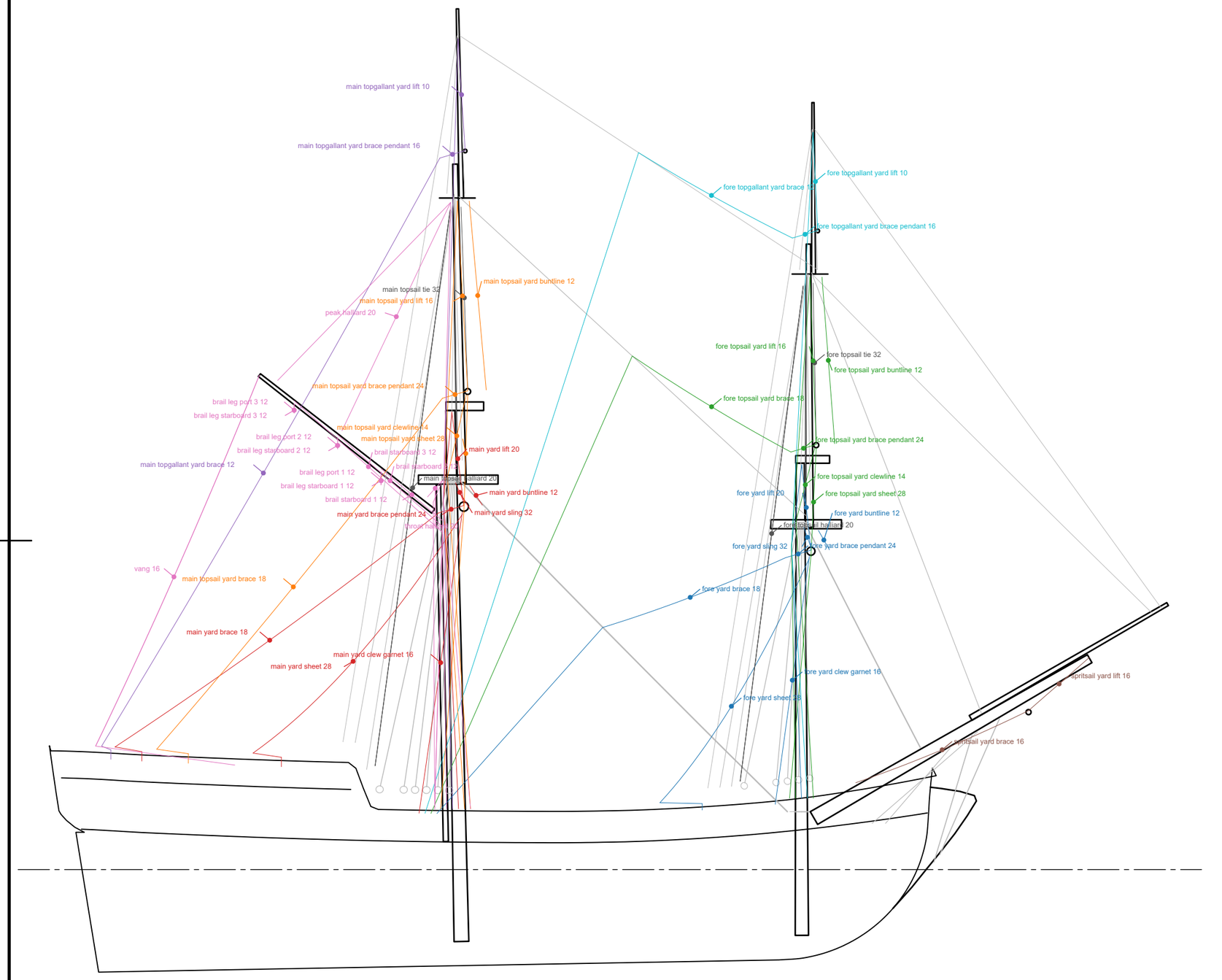
Deadeye and lanyard

Deadeye pair with lanyard, chain plate and channel

Scale	Sheet	Format
1:10	15 / 19	A3, dimensions in metres

Reference model - snow, about 1710 - drawn by script from the model parameters

Side elevation from starboard, sails furled 1:150



Colour = yard or spar the rope works

- fore yard
- fore topsail yard
- fore topgallant yard
- main yard
- main topsail yard
- main topgallant yard
- spritsail yard
- gaff
- other

Numbers after the names: diameter in mm.

Starboard ropes shown; port side mirrored.

Standing rigging in light grey.

1:150

0 1 2 3 4 5 6 7 8 9 10 m

Leads of the reference model (chapter 07); belaying points: belaying plan.

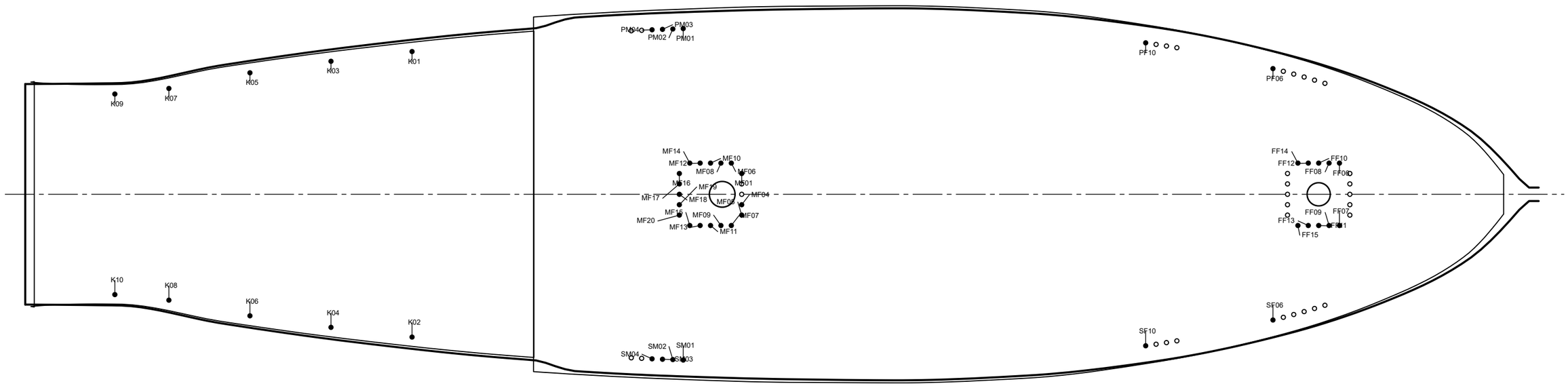
Running rigging

Lifts, braces, clewlines, sheets, buntlines, halliards, ties, gaff ropes

Scale	Sheet	Format
1:150	16 / 19	A3, dimensions in metres

Reference model - snow, about 1710 - drawn by script from the model parameters

Belaying points, plan (reconstruction) 1:80



Pin or kevel - rope belayed (reference model, reconstruction)

FF06 fore topsail yard clewline port
FF07 fore topsail yard clewline starboard
FF08 fore topgallant yard lift port
FF09 fore topgallant yard lift starboard
FF10 fore yard lift port
FF11 fore yard lift starboard
FF12 fore topsail yard lift port
FF13 fore topsail yard lift starboard
FF14 fore topsail yard sheet port
FF15 fore topsail yard sheet starboard
MF01 main topsail yard clewline port
MF04 peak halliard
MF05 main topsail yard clewline starboard
MF06 main topsail yard lift port
MF07 main topsail yard lift starboard
MF08 main yard lift port
MF09 main yard lift starboard
MF10 main topgallant yard lift port
MF11 main topgallant yard lift starboard
MF12 brail port 3
MF13 brail starboard 3
MF14 main topsail yard sheet port
MF15 main topsail yard sheet starboard
MF16 brail port 1
MF17 brail port 2
MF18 throat halliard
MF19 brail starboard 2
MF20 brail starboard 1
PF06 fore yard clew garnet port
PF10 fore yard sheet port
SF06 fore yard clew garnet starboard
SF10 fore yard sheet starboard
PM01 fore yard brace port
PM02 fore topsail yard brace port
PM03 fore topgallant yard brace port
PM04 main yard clew garnet port
SM01 fore yard brace starboard
SM02 fore topsail yard brace starboard
SM03 fore topgallant yard brace starboard
SM04 main yard clew garnet starboard
K01 main yard sheet port
K02 main yard sheet starboard
K03 vang port
K04 vang starboard
K05 main topsail yard brace port
K06 main topsail yard brace starboard
K07 main yard brace port
K08 main yard brace starboard
K09 main topgallant yard brace port
K10 main topgallant yard brace starboard

FF/MF fore/main fife rail, PF/SF/PM/SM pin rails port/starboard fore/main, K kevels; open circles: free pins.

Reconstruction for the reference model; pin rails along the bulwark are a likely anachronism for 1710 (chapter 07).

Belaying plan

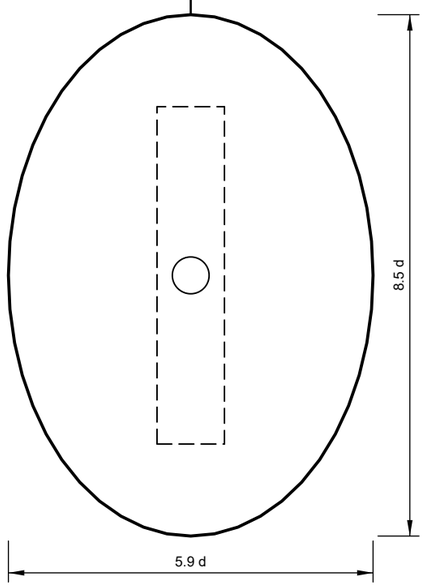
Which rope is belayed on which pin, cleat or kevel

Scale	Sheet	Format
1:80	17 / 19	A3, dimensions in metres

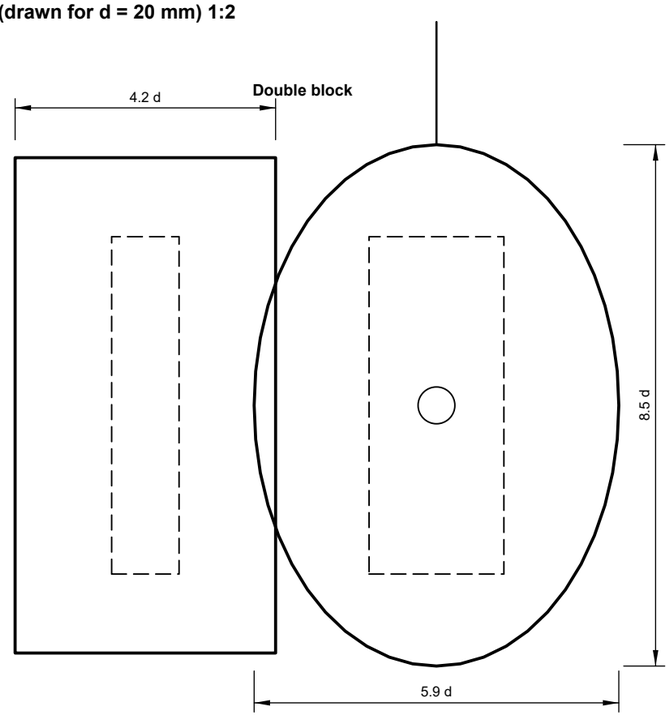
Reference model - snow, about 1710 - drawn by script from the model parameters

Blocks, deadeye and heart - proportions in rope diameters d (drawn for d = 20 mm) 1:2

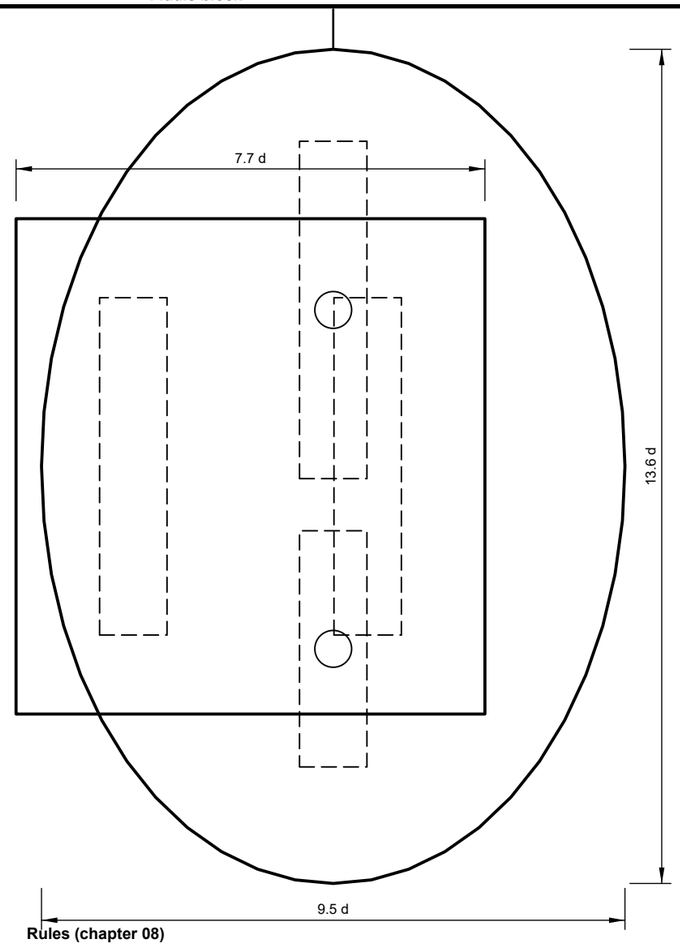
Single block



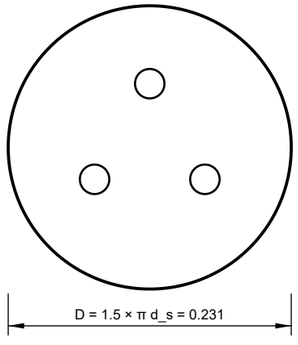
Double block



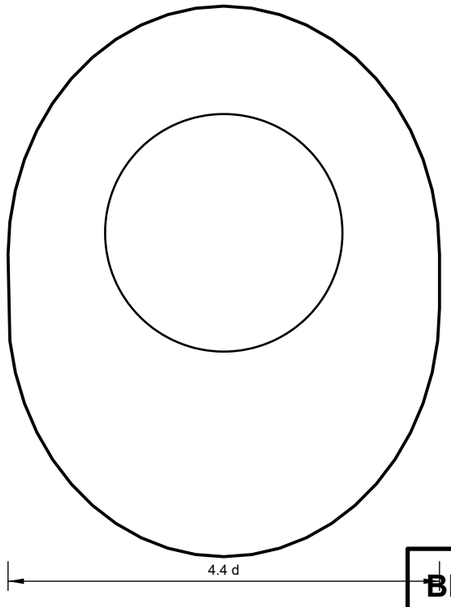
Fiddle block



Deadeye (main shroud d_s = 49 mm), 1:5



Heart (main stay collar, d = 80 mm), 1:5



Rules (chapter 08)

sheave thickness	1.1 d
sheave diameter	5.5 d
block length	8-9 d
block breadth	0.7 × length
block thickness	0.5 × length + 0.4 × length per extra sheave
deadeye diameter	1.5 × circumference of the shroud
deadeye thickness	D/2
heart (our inference)	4.4 d outside, hole 2.4 d

Reference model block lengths: 0.10-0.22 m (lifts 0.14, braces 0.16, topsail halliard 0.20).

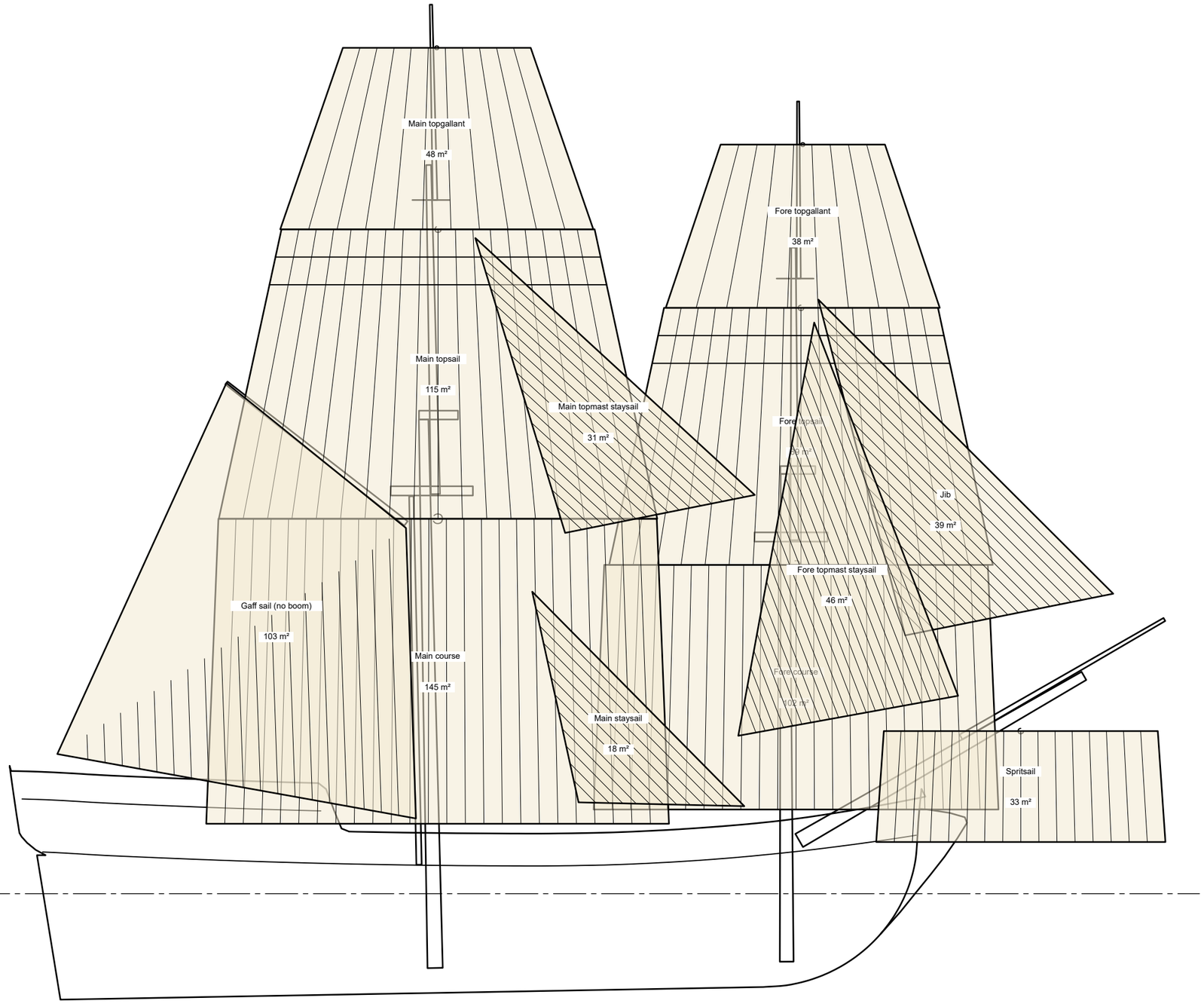
Blocks and deadeyes

Proportions as multiples of the rope diameter

Scale	Sheet	Format
1:2, 1:5	18 / 19	A3, dimensions in metres

Reference model - snow, about 1710 - drawn by script from the model parameters

Sail plan, all sails set (estimated from the spars) 1:150



Sail areas (our inference)

Fore course	101.6 m²
Fore topsail	89.1 m²
Fore topgallant	37.7 m²
Main course	144.8 m²
Main topsail	114.6 m²
Main topgallant	47.9 m²
Spritsail	32.9 m²
Gaff sail (no boom)	102.6 m²
Jib	38.7 m²
Fore topmast staysail	46.3 m²
Main topmast staysail	31.3 m²
Main staysail	18.3 m²
Total	**806 m²**

Head 0.9 × yard; foot = head of the sail below (course: 0.95 × its yard);
yards hoisted to the working position;
courses to 0.3 m above the rail;
cloths 0.61 m, seams every 0.57 m;
topsails 2 reef bands (1710).
Square sails drawn in profile as if braced fore and aft (sail plan convention).

The 3D model carries its sails furled; set sails here are an estimate (chapter 08).

1:150

0 1 2 3 4 5 6 7 8 9 10 m

Sail plan

Outlines computed from the spars; seams, reef bands and area of each sail

Scale	Sheet	Format
1:150	19 / 19	A3, dimensions in metres

Reference model - snow, about 1710 - drawn by script from the model parameters